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Chapter 16

WHAT CAN THE CEREBRAL CORTEX DO BETTER THAN OTHER PARTS OF THE BRAIN?

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INTRODUCTION

One of the most striking results of mammalian evolution is the development of a large cerebral cortex. A cortex or pallium, a planar piece of nerve tissue at the surface of the forebrain, is there in most vertebrates, but its size (relative to the rest of the brain) usually does not reach that in mammals. In all mammals, the cortex is the most impressive subdivision of the brain. In mice, for example, the cortex amounts to about 40% of the brain weight (estimated from measurements on cortical volume; Schüz and Palm, 1989); in humans it reaches nearly 80% (gray and white matter taken together) of the brain weight (estimated from the 76% isocortical volume in Frahm et al., 1982). This raises the question as to the evolutionary advantage of this part of the brain. It can be assumed to be related to the fact that the cerebral cortex is responsible for higher cognitive functions as is evident from neurological deficits in humans.

A comparison among various mammalian species shows that it is mainly the dorsal part of the cortex, the isocortex, that increases along the mammalian evolutionary scale. The surrounding parts, considered to be homologues to most of the reptilian cortex (e.g., Desan, 1988) and often described by the term *allocortex*, remain relatively small. In humans, 96% of the cortical surface area belongs to the isocortex, while in hedgehogs it is only 32% (Filimonov, 1949, quoted in Blinkov and Glezer, 1968). A clue to the understanding of the enlargement of the dorsal part may be the fact that this part of the cortex receives input from all the sensory systems (preprocessed in other parts of the brain). In reptiles, too, there is a small dorsal region, which receives polysensory input (e.g., Hoogland and Vermeulen-Van der Zee, 1988). Thus, when seeking the evolutionary advantage of the enlargement of the cerebral cortex, one may assume that it is connected to the possibility of coordinating the various sensory systems. This is certainly a prerequisite for higher cognitive functions.

The very fact that the cortex is a place where the sensory systems meet, however, is probably not a sufficient explanation for its special development. There are other parts of the brain, where at least some of the sensory systems are represented close to each other, for example, the thalamus or the cerebellum. This leads to the suspicion that, in addition to polysensory input, there is something in the intrinsic connectivity of the cortex that is advantageous to higher cognitive functions.

Indeed, microscopic observations indicate that the various parts of the brain, besides being characterized by their input and output relationship and their relative size, also show characteristic differences in their intrinsic structure. Some parts of the brain, for example, the olfactory bulb or the cerebellum, have so typical a structure that they can be recognized at first sight in the microscope. In other cases more time and practice is required, but it is possible in principle. One may assume that the typical appearance of the various parts of the brain reflects different kinds of connectivity and, consequently, different kinds of information processing.

Thus, with respect to the success of cortical evolution, the question arises: What is the typical connectivity of the cerebral cortex, and what made it so particularly suited for higher cognitive functions? This has been a long-standing question in our group (Braitenberg, 1977, 1978a, 1986; Palm, 1982; Braitenberg and Schüz, 1991, 1998), and the main points are summarized in this chapter.

BASIC CONNECTIVITY OF THE ISOCORTEX

The cortex, like the rest of the brain, consists mainly of nerve cells, the processes of which (axons and dendrites) are interwoven at the highest possible packing density. In the case of the cerebral cortex, this compact mass forms a flat sheet and is separated into a superficial gray matter in which the neurons have their synaptic connections and a white matter that consists of long axonal processes, forming a kind of cable system underneath the gray matter. The thickness of the gray matter does not vary much between species: about 1 mm in mice and about 3 mm in humans. Under $1 mm^2$, the number of neurons is of the order of 10^5 (Rockel, 1980), with some variability between areas or species. The surface area differs, however, by a factor of 1000 between these species. As a consequence, the cortex is folded in larger brains.

The entire isocortex can be viewed as a continuous network despite the fact that it can be subdivided morphologically into different areas. These areas constitute local variations of the common basic connectivity, which is described below. Fibers emanating from the sensory systems radiate into the cortex and project the various sensory surfaces onto separate areas.

The various parts of the brain can usually be characterized by the presence of particular cell types. These are classified mainly by the shape or size of their cell bodies, by the shape and size of their dendritic and axonal ramifications, or by the outlines of their dendrites, which can be smooth, beaded, or spiny.

Typical of the cerebral cortex is the so-called pyramidal cell (Fig. 16.1). Its cell body is often pyramidal in shape, and its dendritic tree is usually elongated, its long axis directed perpendicularly toward the surface of the cortex. The axon is directed in the opposite direction and enters the white matter. The dendrites are densely studded with spiny processes, each of which carries a synapse. About 85% of cortical neurons are pyramidal cells (Peters and Kara, 1985; Braak and Braak, 1986). The other

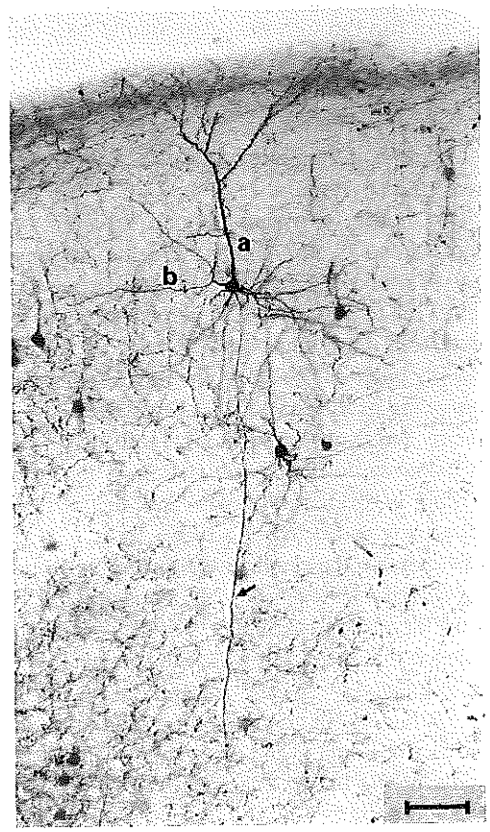

Figure 16.1. Pyramidal cell in the cortex of the mouse, stained by retrograde tracing with biotinylated dextran amine. a, Apical dendrite, reaching the surface of the cortex; b, basal dendrites, emananting from the cell body; arrow, main axon, running toward the white matter. Bar = 50 µm.

neurons in the cortex are loosely interspersed among the pyramidal cells. Their dendrites are smooth, and their axons do not leave the gray matter.

The connectivity of a given part of the brain normally cannot be derived directly from its light or electron microscopical appearance. This is particularly difficult in the cerebral cortex, where, under the microscope, one is confronted with thousands of fibers crossing each other with no obvious order or aim. Combinations of methods are, thus, necessary. Our approach relied on a quantitative assessment of the cell parts (cell bodies, spines, synapses, axons, dendrites) that can be visualized in the microscope with various histological methods (summarized in Braitenberg and Schüz, 1991, 1998). In the next few paragraphs, the most important results are summarized and compared with those from other parts of the brain (see also the review by Nieuwenhuys, 1994).

The neurons in the cortex are mainly connected to other cortical neurons. The axons of the pyramidal cells give off collaterals during their course toward the white matter and connect to neurons in their neighborhood. The axons then leave the gray matter, and most of them run through the white matter to a distant place, where they reenter the gray matter and connect to neurons there. Only a small proportion of the pyramidal cells are true output neurons (probably below 10% in humans) that send their axons to other parts of the central nervous system. Input fibers from the sensory systems and from other parts of the brain to the cortex are also much lower in number than the reentering axons of pyramidal cells.

In the cortex of the mouse, a pyramidal cell receives on average about 8000 synapses on its dendrite and makes about the same number of synapses with its axonal ramifications. About 85% of the synapses connect the neuron to other pyramidal cells, about 11% to nonpyramidal cells, and a few percent of the synapses on the dendrites come from input neurons. A pyramidal cell usually makes only one or a few synpases with each of its postsynaptic neighbors. *Thus, the individual pyramidal cell is connected to thousands of other pyramidal cells.*

The synapses made by the axons of pyramidal cells are of a special morphological type (type I), which is known to be *excitatory* (e.g., Peters and Jones, 1984). When the postsynaptic neighbor is another pyramidal cell, the synapse is usually *located on a dendritic spine* (for review see White, 1989). If it is a nonpyramidal cell, the synapse is located directly on the dendrite or on the cell body. The axons of nonpyramidal cells make inhibitory synapses (e.g., Houser et al., 1984). They are usually located directly on the dendrites (or the cell bodies) of the postsynaptic neurons. There is indirect evidence that the dendritic spines play a role in the regulation of the strength of the synapse located on them (summarized by Horner, 1993).

A glance at the other relatively large parts of the brain, the cerebellum, the striatum, and the thalamus, shows that some of the properties of the cortex also appear in these and perhaps other parts of the brain. The entire set of properties mentioned above, however, is unique to the cortex. This indicates that the role of the cortex must be quite different from that of the other brain parts (Fig. 16.2). For example, the striatum also consists mainly of one type of neuron and, as in the cortex, these neurons are connected among each other. They are, however, connected by inhibitory synapses (for review and functional interpretation, see Wickens, 1993; Plenz and Aertsen, 1996). In the thalamus, the majority of the neurons seem to be excitatory, as in the cortex. These, however, relay the sensory input to the cerebral cortex and do not seem to be directly connected to each other (Steriade et al., 1990; for functional interpretation, see Miller, 1996). Dendritic spines are also a typical feature of neurons in the cerebellum, the Purkinje cells. These spines do not, however, receive synapses from the same type of neurons, but from granular cells. In the cerebellar cortex, there is no intrinsic positive feedback either (Braitenberg et al., 1997). Inhibitory interneurons are present both in the cerebellar cortex and in the thalamus.

THE CEREBRAL CORTEX AND COGNITION

What could be the functional role of the kind of connectivity typical of the cerebral cortex? The answer is that it is particularly well suited for functions that are essential for cognitive processes (viz., for storing, retrieving, and handling correlations).

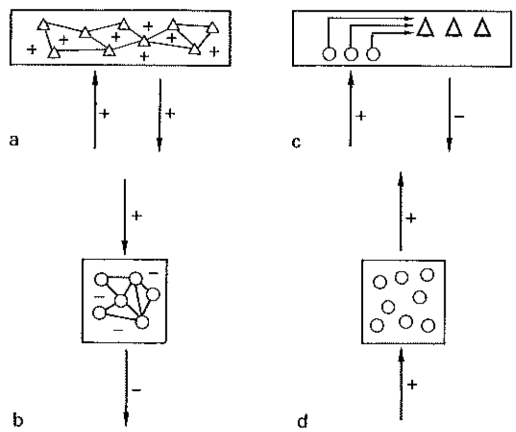

Figure 16.2. Sketches characterizing some aspects of the connectivity of four brain regions. (a) Cerebral cortex. The main input and the output are excitatory. The main cell type forms an excitatory network. (b) Striatum. The input from the cortex is excitatory, the output is inhibitory, and the main cell type forms an inhibitory network. (c) Cerebellar cortex. The input is excitatory, the output inhibitory. The input is relayed by granular cells (round) to the output neurons, the Purkinje cells (triangular), in a typical geometrical arrangement. (d) Thalamic relay nucleus. Input and output are excitatory. The projection neurons do not form a network among themselves.

A mechanism for the storage of correlations was proposed by Hebb (1949) in his theory of cell assemblies, a concept that was taken up again by Legendy (1971) and further elaborated by Braitenberg (1978b). According to this theory, learning is achieved by strengthening the connections between those neurons that are often active together. When a child is exposed to a new object, the various properties of the object, such as its particular color, shape, consistency, and so forth, will be projected onto the cortex by the sensory input systems and will excite a certain set of neurons there. This set will be more or less the same, whenever this or a similar object reappears. If there happen to be connections between members of this set, these connections might be strengthened whenever the neurons fire together. After a while, the connections will have become strong enough to ignite the whole set when only some of its members are activated from outside. By then, the new object will have a representation in the brain, and the concept of it can be evoked relatively independently from the input. Such "cell assemblies" might constitute the units of cognition (Fig. 16.3).

According to this theory, individual neurons of a cell assembly might be part of other cell assemblies representing concepts with similar properties. A train of thoughts could then result from the successive ignition of overlapping cell assemblies. It is difficult to prove this theory, but it turned out to be a very promising starting point whenever trying to pin down cognitive functions to the neuronal level (Braitenberg, 1985; Palm, 1990; Miller and Wickens, 1991; Braitenberg and Pulvermüller, 1992; Braitenberg and Schüz, 1992; Pulvermüller, 1992; Wickelgren, 1992).

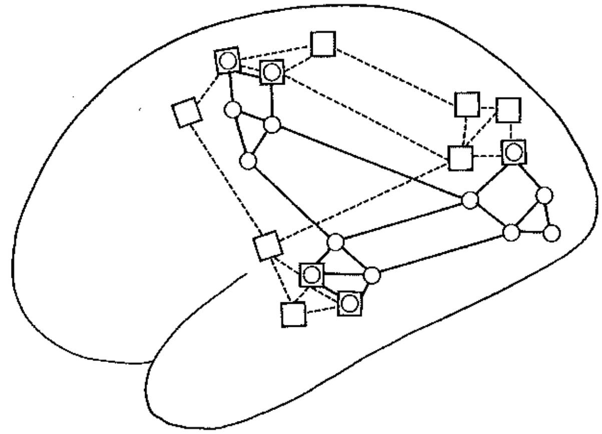

Figure 16.3. Scheme of two overlapping cell assemblies in a human brain. The neurons, indicated by circles or squares, respectively, are representative of neurons in higher (not primary) sensory areas (visual, acoustic, somatosensory).

When seeking the properties required by a piece of nervous tissue for the Hebbian kind of learning, one would end up with the connectivity typical of the cerebral cortex (Table 16.1). The tissue should receive input from the various sensory systems and should be composed of many neurons of the same kind. The neurons should be connected into a network. The synapses between the neurons should be modifiable in strength. This can be assumed to be the case for the synapses between pyramidal cells, most of which are located on dendritic spines. The connections should be excitatory in order to translate correlated activity in the input into correlated activity in the network. The network should be large; the more neurons there are, the more correlations can be detected and stored.

A large number of neurons and excitatory connections between them are also required for the development and maintenance of a train of thoughts. While it is still difficult to imagine the details of long-lasting time-structured events in the cortex such as a coherent train of thoughts, it is even more difficult or almost impossible to imagine such events in other parts of the mammalian brain that are either small or

Table 16.1: Requirements for Cell Assemblies and Structure of the Cortex

Requirements	Structure
Many neurons of the same kind	About 85% pyramidal cells
Connected with each other	Most synapses are between pyramidal cells
Excitatory connections	About 89% type I synapses
Modifiable connections	About 75% synapses on spines
Individual neurons connected to as many others as possible	About 8000 synapses/neuron
Connections between distant regions of the network	Large amount of white matter

characterized by inhibitory interactions. An approach to long-lasting dynamic aspects of cognitive functions is provided by the theory of synfire chains (Abeles, 1991), which is also based on the particular connectivity of the cerebral cortex and which can be well combined with the theory of cell assemblies.

Another important requirement would be that the network, despite its large size, should be richly connected within itself. Connections should also be possible over long distances. Each part of the network should be able to easily communicate with other parts of the network in order to be prepared for all possible constellations of common activity that might appear in the particular environment to which an individual is exposed. A highly specific prewiring between individual neurons is unlikely and would actually defeat the purpose of a learning device. A prewiring that enables individual elements to interact with a large number of other elements seems to be more advantageous.

Although the cerebral cortex is far from being a fully connected network in which each element is connected to every other one, one of the really unique features of the cortex is its rich connectivity within itself. This is evident from the large amount of white matter in larger brains, as well as from the fact that most pyramidal cells connect to thousands of others, both in their vicinity and at a distant place. In the isocortex of the mouse, the connectivity is such that each region is connected to every other by one or a few synaptic steps (Palm, 1984).

In conclusion, the cerebral cortex, like no other part of the brain, has the connectivity required for storing and handling a large number of correlations. It can be assumed to be a particularly efficient learning device in which memory items can be retrieved and combined in ever new ways, allowing for the detection of further and more complex correlations.

COMPARATIVE ASPECTS

Brain Size and Connectivity

The striving for a high degree of connectivity also becomes evident when comparing small and large mammalian brains. A large cortex has the advantage of having a large storage capacity, but it may be difficult for it to maintain the same degree of connectivity of a network with a smaller number of neurons. The attempt to uphold connectivity can be deduced from the following trends:

1. The volume of the cortical white matter increases much more than the volume of the gray matter. In some basal insectivores, the white matter amounts to only 6% of the isocortical volume, whereas in apes and humans it is around 40% (Frahm et al., 1982).

2. In brains with a larger cortex, individual neurons make more synapses on average. This can be deduced from the facts that the density of neurons per cubic millimeter decreases in larger brains (for review, see Jerison, 1973) and

that the density of synapses per cubic millimeter seems to stay constant, as has been shown for hedgehog and monkey (Schüz and Demianenko, 1995).

3. In addition, the average thickness of fibers increases, partially compensating for the longer conduction times that come with the larger distances in larger brains.

All of this cannot, however, fully compensate for the larger size of the cortex. For example, the average number of synapses per neuron may differ by a factor of at most 10 across mammalian species (Schüz and Demianenko, 1995), while the number of neurons in the cortex may differ by a factor of more than 1000. Furthermore, the average thickness of fibers does not increase sufficiently either to compensate for the longer conduction times that come with the increased distances (Jerison, 1991; Schüz and Preißl, 1996). This is suggested by measurements of the thickness of myelinated fibers in the corpus callosum of mouse and monkey (Jerison, 1991; Schüz and Preißl, 1996), as well as by calculations by Ringo et al. (1994), who show that otherwise large brains would need to be much larger than they are. Although individual fibers can become very thick in the monkey (Fig. 16. 4), the bulk of (myelinated) fibers has a quite similar diameter in small and large brains (between 0.5 and 1 μm in mouse and monkey). As pointed out by Ringo (1991), this should lead to a higher degree of local specialization in larger mammalian brains.

Allocortex and Reptilian Cortex

Both the mammalian allocortex and the reptilian cortex differ architectonically from the mammalian isocortex, and a lack of quantitative data on these structures makes it difficult to compare their connectivity with that of the isocortex. There is evidence, however, that both allocortex and reptilian cortex share important properties with the

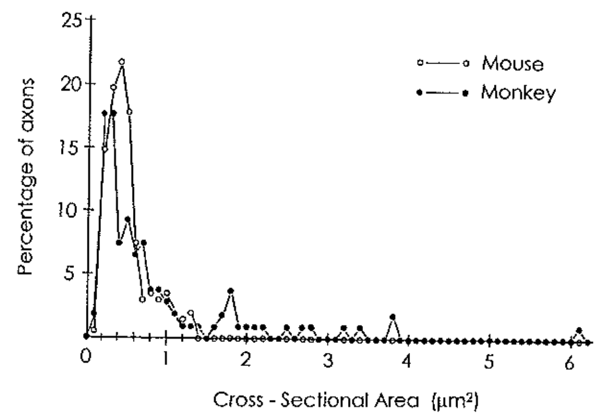

Figure 16.4. Thickness of myelinated axons in the corpus callosum of mouse and monkey. Each symbol corresponds to the relative frequency of axons of a given cross-sectional area. Bin width: 0.1 μm^2. (Same data as in Jerison, 1991; modified from Schüz and Preißl, 1996.)

mammalian isocortex. For example, in the mammalian hippocampus, excitatory connections between pyramidal cells also constitute a large proportion of the connectivity (Braitenberg and Schüz, 1983). In the reptilian cortex, it has been shown that there too the various parts of the cortex connect to each other (Halpern, 1980; Desan, 1988; Hoogland and Vermeulen-Van der Zee, 1988), and there are also cells that resemble the pyramidal cells of the mammalian cortex. They carry spines, and there is evidence for mutual excitation of these neurons (Kriegstein and Connors, 1986; Reiner, 1993).

Thus, the basic requirements for storing correlations in a Hebbian manner are also there in the reptilian cortex. With the elaboration of the acoustic and olfactory system in mammals, which can be assumed to be the result of the invasion of the nocturnal niche by the early mammals (Jerison, 1973), a more detailed picture of the environment reached the cortex. A larger and more elaborated learning device was certainly advantageous. The most important improvement may have been the introduction of the elaborated system of long-range connections (Miller, in press) that is so typical of the mammalian isocortex. The long-range connections are mainly made by the pyramidal cells from the upper cortical layers, and it is that layers which seem to be lacking in the reptilian cortex (Reiner, 1993).

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BRAIN EVOLUTION AND COGNITION

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